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# MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

**Paper Code : IT-705E**

**PUID : 07061 (To be mentioned in the main answer script)**

## ADVANCED DATA COMMUNICATION & CODING

*Time Allotted* : 3 Hours | *Full Marks* : 70

*The figures in the margin indicate full marks.*

*Candidates are required to give their answers in their own words as far as practicable.*

### GROUP – A

### ( Multiple Choice Type Questions )

1. Choose the correct alternatives for any *ten* of the following : $10 \times 1 = 10$

   i) A single frame in GSM frame structure consists of
   a) 8 time slots b) 10 time slots
   c) 12 time slots d) 16 time slots.

   ii) Maximum unit of BER allowed in optical communication system for faithful digital transmission is
   a) $10^{-19}$ b) $10^{-9}$
   c) $10^{9}$ d) $10^{19}$.

   iii) Optical bandwidth is always
   a) greater than the electrical bandwidth
   b) less than the electrical bandwidth
   c) equal than the electrical bandwidth
   d) square than the electrical bandwidth.



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iv) The interface between MSC and BSC is
   a) radio interface  b) abis interface
   c) A-interface  d) SS7.

v) Which multiplexing technique transmit digital signal ?
   a) FDM  b) TDM
   c) WDM  d) Both (a) and (b).

vi) The uplink frequency of C-band is
   a) 4 GHz  b) 6 GHz
   c) 8 GHz  d) none of these.

vii) In amplitude phase keying each phase vector is separated by
   a) 90  b) 0
   c) 45  d) 180.

viii) ..... manages the switching function in GSM.
   a) BSS  b) NSS
   c) OSS  d) MSC.

ix) Which of the following subsystem provides radio transmission between mobile station and MSC ?
   a) BSS  b) NSS
   c) OSS  d) BSC.

x) In PCM encoding, quantization level varies as a function of
   a) Frequency
   b) Amplitude
   c) Square of frequency
   d) Square of amplitude.

xi) Adaptive DPCM is used to
   a) increase bandwidth
   b) decrease bandwidth
   c) increase SNR
   d) none of these.



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xii) The down link frequency in the C band transponder is

a) 6 GHz b) 4 GHz
c) 14 GHz d) 11 GHz.

## GROUP – B
### ( Short Answer Type Questions )

Answer any *three* of the following. 3 × 5 = 15

2. Draw and explain the schematic diagram of an optical communication system.

3. a) Write the operating frequency of GSM for uplink and downlink channels.
   b) Define the following terms with their functionality : IMSI, IMEI, T-IMSI.

4. a) What is the difference between soft handoff and hard handoff ?
   b) What are the advantages and disadvantages of I-TCP ? http://www.makaut.com

5. A PCM system uses a uniform quantizer followed by an n-bit encoder. Show that the r.m.s. signal to quantization ratio is approximately given as ( 1·76 + 6 $n$ ) dB.

6. a) What do you mean by ISI ?
   b) State and explain Nyquist Criterion for Zero ISI.

## GROUP – C
### ( Long Answer Type Questions )

Answer any *three* of the following. 3 × 15 = 45

7. a) What are step-index and graded-index optical fibres ?
   b) Explain why the performance of multimode graded-index fibre is improved over multimode step-index fibre.
   c) Briefly explain the architecture of Optical Transport Network.
   d) What is bit interleaving ?



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8. a) With neat block diagram explain the generation and detection of the BPSK signal.
   b) What are the limitations of DM ? How it can be overcome ?
   c) A binary data is transmitted at a rate of $10^6$ bit/sec over a channel having a BW of 3 MHz. Assume that the noise power spectral density at the receiver is $N_0/2 = 10^{-10}$ W/Hz. Determine the average carrier amplitude required at the receiver input for coherent PSK and DPSK signaling schemes to maintain $P_e < 10^{-4}$.

9. Explain the generation & detection of the BPSK signal with neat block diagram.

10. a) Compare the characteristic features of FDMA, TDMA and CDMA mechanisms.
    b) Define the term 'EIRP'.
    c) A satellite downlink at 10 GHz operates with a transmit power of 5 W and an antenna gain of 48·2 dB. Estimate the FIRP in dBW.

11. a) Explain Pulse Code Modulation technique with neat block diagram.
    b) Give a sine wave of frequency $f_m$ and amplitude $A_m$ applied to delta modulator having step size $\Delta$. Show that the slope overload distortion will occur if $A_m > \Delta/2\pi f_m T_s$. Where $T_s$ is the sample period.
    c) Explain the importance of optical switches in networks.